

## CHAPTER 4

### DEVELOPMENT

This chapter discusses about the development part in this project. The development phases comprise creating a prototype of the ZAKATEASE: A Web-Based Zakat Application with QR Code Integration. This chapter also discussed the project requirements, which included hardware, software, and user interface screenshots, sample coding screenshot and database design for the ZAKATEASE: A Web-Based Zakat Application with QR Code Integration. The development part involves the process of building the web-based zakat application with QR code integration using the appropriate tools and software. The process includes the activities front-end, back-end and database development. Each of activities using tools or techniques which are Visual Studio Code for both front-end and back-end development database development using MySQL of phpMyAdmin.

#### 4.1 Hardware Requirements

**Table 4.1** shows that the main hardware utilised in designing and developing ZAKATEASE: A Web-Based Zakat Application with QR Code Integration is an Asus K551LN with a CPU Intel(R) Core (TM) i7-4500U CPU @ 1.80GHz 2.40 GHz, Random Access Memory (RAM) 12.0 GB, and a 64-bit operating system.

**Table 4.1** Hardware requirements

| MODEL | COMPONENTS       | SPECIFICATION                                      |
|-------|------------------|----------------------------------------------------|
|       | Processor        | Intel(R) Core (TM) i7-4500U CPU @ 1.80GHz 2.40 GHz |
|       | Operating System | Windows 10                                         |

|             |                            |                                              |
|-------------|----------------------------|----------------------------------------------|
| Asus K551LN | Random Access Memory (RAM) | 12.0 GB                                      |
|             | System Type                | 64-bit operating system, x64-based processor |

## 4.2 Software Requirements

Appropriate software is required to develop the ZAKATEASE: A Web-Based Zakat Application with QR Code Integration. The following **Table 4.2** shows the list of software used for this project.

**Table 4.2** Software requirements

| Software/operating system | Description                                                                                                        |
|---------------------------|--------------------------------------------------------------------------------------------------------------------|
| Microsoft Word            | Microsoft Word is used to write the thesis of the project.                                                         |
| Figma                     | Figma is used to design the user interface in design phase.                                                        |
| Draw.io                   | Draw.io is used to create diagrams such as Entity Relationship Diagram (ERD), Use Case and diagram for each phase. |
| Visual Studio Code        | VS code is used to develop the web-based zakat application in develop phase.                                       |
| Laravel                   | Laravel is a framework used to develop the web-based zakat application along with VS code.                         |
| Mendeley                  | Mendeley is used to cite previous papers for literature reviews.                                                   |

|            |                                                               |
|------------|---------------------------------------------------------------|
| XAMPP      | XAMPP is used for providing useful tools which is phpMyAdmin. |
| phpMyAdmin | phpMyAdmin is used for administration of MySQL databases.     |

### 4.3 Requirements Analysis

In order to meet the user requirements, the user need are crucial to be determined in developing the ZAKATEASE: A Web-Based Zakat Application with QR Code Integration. Both functional and non-functional requirements are required. A functional requirement defines a system and the component of ZAKATEASE: A Web-Based Zakat Application with QR Code Integration while non-functional requirements define a quality attribute used in the ZAKATEASE: A Web-Based Zakat Application with QR Code Integration.

#### 4.3.1 Functional Requirements (user kene interact dgn system)

The Functional requirement of web-based application outline the specific functionalities and features required to meet user needs. These functional requirements provide clarity and guidance in development process. The following is the functional requirement for the project.

- **User Registration and authentication.**

Users can create the account for in the application which guarantees dependable and safe login, registration, and logout procedures.

- **Users fill in the form and save.**

Users which are student or staff are required to fill in the form of the information related with zakat application. After completing the form, the user proceeds with the saving process by clicking the "Save" button.

- **Calculator functionality.**

Users able to perform the calculator functionality to check the eligibility whether the user are qualifying for make the zakat application.

- **QR Code**

- **Profile update**

#### **4.3.2 Non-functional Requirement**

Non-functional Requirement explains the proper behaviour of a system, as opposed to what it should accomplish and provides the precise tasks that a system must complete. In essence, non-functional requirements are characteristics that the system needs to have. The following are the non-functional requirement of this project.

- **Usability**

The application is easy to understand and navigate to other page, providing a positive user experience with user friendly interface even for differing degrees of technical proficiency.

- **Compatibility**

The application is compatible for other web browser and devices.

- **Scalability**

The application maintains responsiveness and accommodate an increasing amount of data and users.

- **Performance**

Higher reaction times and a seamless, effective user experience are guaranteed by the application.

## 4.4 Database Design

The database design for the web-based consisting of 14 tables that efficiently organize and manage all relevant data. These tables include accommodation\_information, applicant\_information, applicant\_statement, depend\_head\_fam, details\_of\_aid, emails, financial\_information, interview\_details, interview\_feedback, notifications, parent\_details, registered\_user, sibling\_details, and zakat\_application\_dates as shown in Figure ?? . Each table has a unique function, ensuring that all components of the zakat application process are properly recorded and easily accessible.

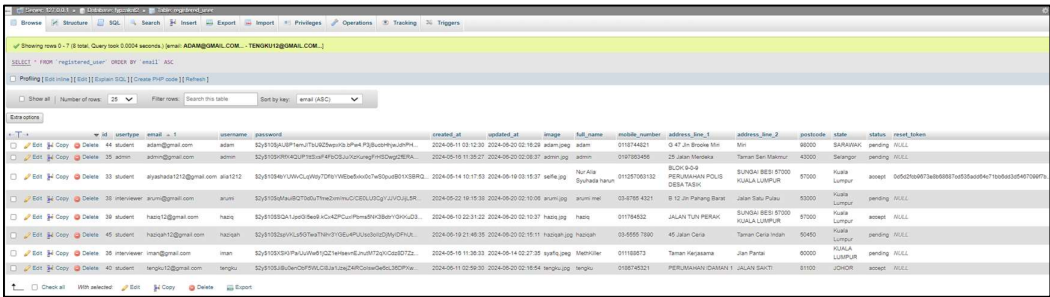
| Table                                              | Action                                      | Rows | Type   | Collation          | Size      | Overhead |
|----------------------------------------------------|---------------------------------------------|------|--------|--------------------|-----------|----------|
| <input type="checkbox"/> accommodation_information | ★ Browse Structure Search Insert Empty Drop | 2    | InnoDB | utf8mb4_general_ci | 32.0 KiB  | -        |
| <input type="checkbox"/> applicant_information     | ★ Browse Structure Search Insert Empty Drop | 3    | InnoDB | utf8mb4_general_ci | 32.0 KiB  | -        |
| <input type="checkbox"/> applicant_statement       | ★ Browse Structure Search Insert Empty Drop | 3    | InnoDB | utf8mb4_general_ci | 32.0 KiB  | -        |
| <input type="checkbox"/> depend_head_fam           | ★ Browse Structure Search Insert Empty Drop | 3    | InnoDB | utf8mb4_general_ci | 32.0 KiB  | -        |
| <input type="checkbox"/> details_of_aid            | ★ Browse Structure Search Insert Empty Drop | 3    | InnoDB | utf8mb4_general_ci | 32.0 KiB  | -        |
| <input type="checkbox"/> emails                    | ★ Browse Structure Search Insert Empty Drop | 6    | InnoDB | utf8mb4_general_ci | 32.0 KiB  | -        |
| <input type="checkbox"/> financial_information     | ★ Browse Structure Search Insert Empty Drop | 3    | InnoDB | utf8mb4_general_ci | 32.0 KiB  | -        |
| <input type="checkbox"/> interview_details         | ★ Browse Structure Search Insert Empty Drop | 3    | InnoDB | utf8mb4_general_ci | 64.0 KiB  | -        |
| <input type="checkbox"/> interview_feedback        | ★ Browse Structure Search Insert Empty Drop | 2    | InnoDB | utf8mb4_general_ci | 64.0 KiB  | -        |
| <input type="checkbox"/> notifications             | ★ Browse Structure Search Insert Empty Drop | 16   | InnoDB | utf8mb4_general_ci | 32.0 KiB  | -        |
| <input type="checkbox"/> parent_details            | ★ Browse Structure Search Insert Empty Drop | 3    | InnoDB | utf8mb4_general_ci | 32.0 KiB  | -        |
| <input type="checkbox"/> registered_user           | ★ Browse Structure Search Insert Empty Drop | 9    | InnoDB | utf8mb4_general_ci | 64.0 KiB  | -        |
| <input type="checkbox"/> sibling_details           | ★ Browse Structure Search Insert Empty Drop | 11   | InnoDB | utf8mb4_general_ci | 32.0 KiB  | -        |
| <input type="checkbox"/> zakat_application_dates   | ★ Browse Structure Search Insert Empty Drop | 1    | InnoDB | utf8mb4_general_ci | 16.0 KiB  | -        |
| 14 tables                                          | Sum                                         | 68   | InnoDB | utf8mb4_general_ci | 528.0 KiB | 0 B      |

Figure ?? List of tables

### 4.4.1 Database Tables

The registered\_user database is critical in the web-based zakat application because it manages user accounts and ensures safe authentication. Each user is assigned a unique id and classified by usertype (student, admin, and interviewer). Email addresses (email) are indexed for easy administration, while usernames and encrypted passwords ensure safe access. Timestamps (created\_at, updated\_at) track account activity, while optional profile image uploads personalise user profiles. Essential facts such as full\_name, mobile\_number, and address lines (address\_line\_1, address\_line\_2, postcode, state) improve user profiles. Account status is for provide the status of

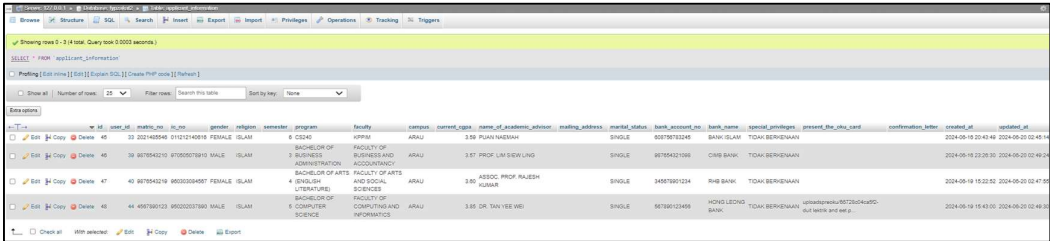
application for each user and reset\_token feature assures secure password recovery.



| id | username    | password              | email                 | created_at          | updated_at          | image        | full_name              | mobile_number | address_line_1                       | address_line_2     | postcode | state        | status  | reset_token                                     |
|----|-------------|-----------------------|-----------------------|---------------------|---------------------|--------------|------------------------|---------------|--------------------------------------|--------------------|----------|--------------|---------|-------------------------------------------------|
| 44 | student     | adam                  | adam@gmail.com        | 2024-08-11 02:12:32 | 2024-08-22 02:18:28 | adam.jpg     | Adam                   | 0118744821    | G 47 Jn Brooke Mtn                   | Mn                 | 88000    | SARAWAK      | pending | NULL                                            |
| 35 | admin       | admin                 | admin@gmail.com       | 2024-08-19 11:38:27 | 2024-08-22 02:18:27 | admin.jpg    | Admin                  | 0197894988    | 25 Jalan Merdeka                     | Taman Sri Mahkota  | 42000    | Selangor     | pending | NULL                                            |
| 33 | student     | ayyasaia121@gmail.com | ayyasaia121@gmail.com | 2024-08-14 10:17:53 | 2024-08-19 10:17:53 | ayyasaia.jpg | Nur Aina Syahida Hanis | 011287580132  | BLD 4-8-B PERUMAHAN POLIS DESA TASEK | SUNGAI BEB 97000   | 87000    | Kuala Lumpur | accept  | 045629a9873a8b55817a5d5b5d4671a0d5d3d45d73d8f7a |
| 38 | interviewer | ayumi@gmail.com       | ayumi@gmail.com       | 2024-08-22 02:18:38 | 2024-08-22 02:18:38 | ayumi.jpg    | ayumi mal              | 03-4785 4321  | 9 12 Jln Parang Barat                | Jalan Sau, Putau   | 83000    | Kuala Lumpur | pending | NULL                                            |
| 39 | student     | hazqz                 | hazqz@gmail.com       | 2024-08-10 22:31:22 | 2024-08-20 02:10:37 | hazqz.jpg    | hazqz                  | 011794552     | JALAN TUN PERAK                      | SUNGAI BEB 97000   | 87000    | Kuala Lumpur | accept  | NULL                                            |
| 45 | student     | hazqz12@gmail.com     | hazqz12@gmail.com     | 2024-08-19 12:48:35 | 2024-08-20 02:10:11 | hazqz.jpg    | hazqz                  | 03-8555 7890  | 45 Jalan Cerdas                      | Taman Cerdas Indah | 50450    | Kuala Lumpur | pending | NULL                                            |
| 38 | interviewer | mang@gmail.com        | mang@gmail.com        | 2024-08-19 11:38:33 | 2024-08-14 12:27:58 | syafiq.jpg   | Mahdzir                | 011188873     | Taman berkecambah                    | Jalan Partak       | 80000    | Kuala Lumpur | pending | NULL                                            |
| 40 | student     | linggu                | linggu12@gmail.com    | 2024-08-11 02:59:37 | 2024-08-20 02:16:54 | linggu.jpg   | linggu                 | 0188748321    | PERUMAHAN DAMANS 1                   | JALAN SAKTI        | 81100    | Johor        | accept  | NULL                                            |

Figure ?? Table registered\_user

The applicant\_information table holds important information regarding applicants. Each entry has an id and a user\_id that point to their user profile. Key personal information is captured, including matric\_no, ic\_no, gender, and religion. Academic characteristics such as semester, programme, faculty, and campus are recorded, in addition to current\_cgpa and name\_of\_academic\_advisor. Contact information (mailing\_address), marital status, and financial information (account\_number, bank\_name) are all controlled. Additional factors include special\_privileges, presenting\_the\_oku\_card, and a confirmation letter. Timestamps (created\_at, updated\_at) monitor application updates, allowing for more effective applicant administration and decision-making.



| id | user_id | matric_no | ic_no        | gender | religion | semester | program          | faculty                                | campus | current_cgpa | name_of_academic_advisor   | mailing_address | marital_status | bank_account_no  | bank_name       | special_privileges                | present_the_oku_card | confirmation_letter | created_at          | updated_at          |
|----|---------|-----------|--------------|--------|----------|----------|------------------|----------------------------------------|--------|--------------|----------------------------|-----------------|----------------|------------------|-----------------|-----------------------------------|----------------------|---------------------|---------------------|---------------------|
| 45 | 33      | 202145548 | 0112145816   | FEMALE | ISLAM    | 8        | CEAS             | APPM                                   | ARAU   | 3.89         | PURNAHENDAH                | SHULE           | 807870248      | BANK ISLAM       | TOAH BERKEKAMAN |                                   |                      |                     | 2024-08-19 20:43:48 | 2024-08-22 02:45:14 |
| 46 | 39      | 867054219 | 870050078910 | MALE   | ISLAM    | 3        | BUSINESS         | FACULTY OF BUSINESS AND ADMINISTRATION | ARAU   | 3.87         | PROF. LIM SEW LING         | SHULE           | 80788421088    | CRB BANK         | TOAH BERKEKAMAN |                                   |                      |                     | 2024-08-19 22:38:10 | 2024-08-20 02:45:29 |
| 47 | 45      | 867054219 | 89000004807  | FEMALE | ISLAM    | 4        | ENGLISH          | FACULTY OF ARTS AND SOCIAL LITERATURE  | ARAU   | 3.86         | ASSOC. PROF. RAHMAD ALBARR | SHULE           | 34857881224    | RHB BANK         | TOAH BERKEKAMAN |                                   |                      |                     | 2024-08-19 15:32:55 | 2024-08-20 02:47:45 |
| 48 | 44      | 408789123 | 950200337890 | MALE   | ISLAM    | 6        | COMPUTER SCIENCE | FACULTY OF COMPUTING AND INFORMATICS   | ARAU   | 3.85         | DR. TAN YEE WEI            | SHULE           | 80788123485    | HONG LECTRA BANK | TOAH BERKEKAMAN | apakahsaya8773dca4a7d3a4b8a9d4e4a |                      |                     | 2024-08-19 16:43:02 | 2024-08-20 02:48:58 |

Figure ?? Table applicant\_information



| id | user_id | matric_no | ic_no        | gender | religion | semester | program          | faculty                               | campus | current_cgpa | name_of_academic_advisor   | mailing_address | marital_status | bank_account_no  | bank_name       | special_privileges                | present_the_oku_card | confirmation_letter | created_at          | updated_at          |
|----|---------|-----------|--------------|--------|----------|----------|------------------|---------------------------------------|--------|--------------|----------------------------|-----------------|----------------|------------------|-----------------|-----------------------------------|----------------------|---------------------|---------------------|---------------------|
| 49 | 45      | 867054219 | 89000004807  | FEMALE | ISLAM    | 4        | ENGLISH          | FACULTY OF ARTS AND SOCIAL LITERATURE | ARAU   | 3.86         | ASSOC. PROF. RAHMAD ALBARR | SHULE           | 34857881224    | RHB BANK         | TOAH BERKEKAMAN |                                   |                      |                     | 2024-08-19 15:32:55 | 2024-08-20 02:47:45 |
| 50 | 44      | 408789123 | 950200337890 | MALE   | ISLAM    | 6        | COMPUTER SCIENCE | FACULTY OF COMPUTING AND INFORMATICS  | ARAU   | 3.85         | DR. TAN YEE WEI            | SHULE           | 80788123485    | HONG LECTRA BANK | TOAH BERKEKAMAN | apakahsaya8773dca4a7d3a4b8a9d4e4a |                      |                     | 2024-08-19 16:43:02 | 2024-08-20 02:48:58 |

Figure ?? Table parent\_details

Showing rows 0 - 3 (4 total, Query took 0.0003 seconds.)

SELECT \* FROM `parent\_details`

Number of rows: 25 Filter rows: Search this table Sort by key: None

|                                                                                             | id | city_of_residence | head_of_family | working_status | working_adults | unemployed_adults | ipt_students | school_children | children_5_years_and_under | applicant_id |
|---------------------------------------------------------------------------------------------|----|-------------------|----------------|----------------|----------------|-------------------|--------------|-----------------|----------------------------|--------------|
| <input type="checkbox"/> Edit <input type="checkbox"/> Copy <input type="checkbox"/> Delete | 8  | WP KUALA LUMPUR   | 1 YES          |                | 3              | 1                 | 1            | 0               | 0                          | 45           |
| <input type="checkbox"/> Edit <input type="checkbox"/> Copy <input type="checkbox"/> Delete | 9  | WP KUALA LUMPUR   | 1 YES          |                | 3              | 2                 | 1            | 0               | 0                          | 46           |
| <input type="checkbox"/> Edit <input type="checkbox"/> Copy <input type="checkbox"/> Delete | 10 | KOTA KINABALU     | 1 YES          |                | 4              | 3                 | 5            | 1               | 0                          | 47           |
| <input type="checkbox"/> Edit <input type="checkbox"/> Copy <input type="checkbox"/> Delete | 11 | KOTA BHARU        | 1 YES          |                | 3              | 3                 | 1            | 4               | 2                          | 48           |

Figure ?? Table depend\_head\_fam

Showing rows 0 - 4 (5 total, Query took 0.0004 seconds.)

SELECT \* FROM `depend\_head\_fam`

Number of rows: 25 Filter rows: Search this table Sort by key: None

|                                                                                             | id | application_id | name                              | relationship  | age | marital_status | working_status | position          | employer | add_info          |
|---------------------------------------------------------------------------------------------|----|----------------|-----------------------------------|---------------|-----|----------------|----------------|-------------------|----------|-------------------|
| <input type="checkbox"/> Edit <input type="checkbox"/> Copy <input type="checkbox"/> Delete | 46 | 45             | NUR AMIERAH SHAQIRAH BINTI HARUN  | Older Sister  | 31  | SINGLE         | EMPLOYED       | TECHNICIAN        | COCHLEAR | NONE              |
| <input type="checkbox"/> Edit <input type="checkbox"/> Copy <input type="checkbox"/> Delete | 47 | 45             | MUHAMMAD AMIRUL HAQIMIE BIN HARUN | Older Brother | 28  | SINGLE         | EMPLOYED       | ENGINEER          | SAMSUNG  | NONE              |
| <input type="checkbox"/> Edit <input type="checkbox"/> Copy <input type="checkbox"/> Delete | 48 | 46             | MOHAMAD FAIZ                      | Older Brother | 45  | SINGLE         | EMPLOYED       | MARKETING MANAGER | PETRONAS | NONE              |
| <input type="checkbox"/> Edit <input type="checkbox"/> Copy <input type="checkbox"/> Delete | 49 | 47             | ZULKIFLI RAZAK                    | Older Brother | 34  | MARRIED        | RETIRED        | N/A               | N/A      | NONE              |
| <input type="checkbox"/> Edit <input type="checkbox"/> Copy <input type="checkbox"/> Delete | 50 | 48             | MOHD YUSOF                        | Older Brother | 43  | MARRIED        | SELF-EMPLOYED  | BUSINESS OWNER    | SELF     | RUNS A RESTAURANT |

Figure ?? Table sibling\_details

Showing rows 0 - 3 (4 total, Query took 0.0017 seconds.)

SELECT \* FROM `sibling\_details`

Number of rows: 25 Filter rows: Search this table Sort by key: None

|                                                                                             | id | application_id | sponsorship_loan | reason                 | source    | sponsorship_amount | sponsorship_balance | tuition_fee | fee_status |
|---------------------------------------------------------------------------------------------|----|----------------|------------------|------------------------|-----------|--------------------|---------------------|-------------|------------|
| <input type="checkbox"/> Edit <input type="checkbox"/> Copy <input type="checkbox"/> Delete | 4  | 45             | No               |                        | TESP MARA | 4800.00            | 0.00                | 800.00      | Paid       |
| <input type="checkbox"/> Edit <input type="checkbox"/> Copy <input type="checkbox"/> Delete | 5  | 46             | Yes              |                        | MARAA     | 4800.00            | 0.00                | 800.00      | Paid       |
| <input type="checkbox"/> Edit <input type="checkbox"/> Copy <input type="checkbox"/> Delete | 6  | 47             | No               | dont want to have loan | NULL      | NULL               | NULL                | 800.00      | Paid       |
| <input type="checkbox"/> Edit <input type="checkbox"/> Copy <input type="checkbox"/> Delete | 7  | 48             | No               | t35r                   | NULL      | NULL               | NULL                | 345.00      | Not Paid   |

Figure ?? Table financial\_information

Server: 127.0.0.1 » Database: fypzakat2 » Table: accommodation\_information

Showing rows 0 - 3 (4 total, Query took 0.0003 seconds.)

SELECT \* FROM `accommodation\_information`

Profiling [ Edit inline ] [ Edit ] [ Explain SQL ] [ Create PHP ]

Show all | Number of rows: 25 | Filter rows: Search this table | Sort by key: None

Extra options

|                          |                  | id | application_id | accommodation_type | accommodation_fee | payment_rate | vehicle | vehicle_type | vehicle_model |
|--------------------------|------------------|----|----------------|--------------------|-------------------|--------------|---------|--------------|---------------|
| <input type="checkbox"/> | Edit Copy Delete | 3  | 45             | college            | paid              | 200.00       | no      |              |               |
| <input type="checkbox"/> | Edit Copy Delete | 4  | 46             | college            | free              | 133.00       | no      |              |               |
| <input type="checkbox"/> | Edit Copy Delete | 5  | 47             | college            | free              | 100.00       | no      |              |               |
| <input type="checkbox"/> | Edit Copy Delete | 6  | 48             | college            | paid              | 300.00       | no      |              |               |

Check all | With selected: Edit Copy Delete Export

Figure ?? Table accommodation\_information

Server: 127.0.0.1 » Database: fypzakat2 » Table: details\_of\_aid

Showing rows 0 - 3 (4 total, Query took 0.0003 seconds.)

SELECT \* FROM `details\_of\_aid`

Profiling [ Edit inline ] [ Edit ] [ Explain SQL ] [ Create PHP code ] [ Refresh ]

Show all | Number of rows: 25 | Filter rows: Search this table | Sort by key: None

Extra options

|                          |                  | detail_id | application_id | zakat_received | zakat_year | zakat_semester | zakat_amount | zakat_purpose       | zakat_documents                                       |
|--------------------------|------------------|-----------|----------------|----------------|------------|----------------|--------------|---------------------|-------------------------------------------------------|
| <input type="checkbox"/> | Edit Copy Delete | 6         | 45             | no             | 0          | 0              | 0.00         | FINANCIAL CRISIS    | NULL                                                  |
| <input type="checkbox"/> | Edit Copy Delete | 7         | 46             | no             | 0          | 0              | 0.00         | FINANCIAL SUPPORT   | uploadzakatdoc/duit lektrik and eet.pdf               |
| <input type="checkbox"/> | Edit Copy Delete | 8         | 47             | no             | 0          | 0              | 0.00         | FINANCIAL CRISIS    | uploadzakatdoc/NUR ALIA SYUHADA BINTI HARUN_202148... |
| <input type="checkbox"/> | Edit Copy Delete | 9         | 48             | no             | 0          | 0              | 0.00         | EDUCATIONAL SUPPORT | uploadzakatdoc/ENT ALIA CASE STUDY.pdf                |

Check all | With selected: Edit Copy Delete Export

Figure ?? Table details\_of\_aid



Server: 127.0.0.1 » Database: fypzakat2 » Table: applicant\_statement

☐ Show all | Number of rows: 25 | Filter rows: Search this table

Extra options

|                          |                  | statement_id | application_id | statement |
|--------------------------|------------------|--------------|----------------|-----------|
| <input type="checkbox"/> | Edit Copy Delete | 7            | 45             | 1         |
| <input type="checkbox"/> | Edit Copy Delete | 9            | 46             | 1         |
| <input type="checkbox"/> | Edit Copy Delete | 10           | 47             | 1         |

☐ Check all | With selected: Edit Copy Delete Export

Figure ?? Table applicant\_statement

Server: 127.0.0.1 » Database: fypzakat2 » Table: interview\_details

☐ Show all | Number of rows: 25 | Filter rows: Search this table | Sort by key: None

Extra options

|                          |                  | id | application_id | interview_date | interview_time | interview_location          | interviewer_id | additional_notes                                       | interview_status | qr_code_path                 | notification_sent | email_sent |
|--------------------------|------------------|----|----------------|----------------|----------------|-----------------------------|----------------|--------------------------------------------------------|------------------|------------------------------|-------------------|------------|
| <input type="checkbox"/> | Edit Copy Delete | 21 | 45             | 2024-05-20     | 04:02:00       | Bilik Penyarah alia syuhada | 36             | Congratulation, you are pass to go to the interview... | Completed        | qr_codes/interview_qr_45.jpg | 0                 | 0          |
| <input type="checkbox"/> | Edit Copy Delete | 25 | 46             | 0000-00-00     | 00:00:00       |                             | 36             | NULL                                                   | pending          |                              | 0                 | 0          |
| <input type="checkbox"/> | Edit Copy Delete | 26 | 47             | 2024-05-05     | 00:00:00       | bilik lecturer              | 36             | Congratulation, you are pass to go to the interview... | pending          | qr_codes/interview_qr_47.jpg | 1                 | 0          |

☐ Check all | With selected: Edit Copy Delete Export

Figure ?? Table interview\_details

Server: 127.0.0.1 » Database: fypzakat2 » Table: interview\_feedback

Showing rows 0 - 0 (1 total. Query took 0.0093 seconds)

SELECT \* FROM 'interview\_feedback'

Profiling [Edit view] [Edit] [Explain SQL] [Create PHP code] [Refresh]

UPDATE 'interview\_feedback' SET 'appraisers\_comments' = 'I support the application' WHERE 'interview\_feedback' . id = 30;

[Edit view] [Edit] [Create PHP code]

☐ Show all | Number of rows: 25 | Filter rows: Search this table

Extra options

|                          |                  | id | application_id | interview_id | only_on_zakat | other_income | additional_documents | selected_options | zakat_eligibility | zakat_eligibility | zakat_amount | zakat_puran               | appraisers_comments | assaf_course_needed | course_recommendations | created_at          | feedback_submitted |
|--------------------------|------------------|----|----------------|--------------|---------------|--------------|----------------------|------------------|-------------------|-------------------|--------------|---------------------------|---------------------|---------------------|------------------------|---------------------|--------------------|
| <input type="checkbox"/> | Edit Copy Delete | 30 | 45             | 21           | yes           | 34           |                      | orphans          | eligible          | partial_financing | RM1000       | I support the application | no                  |                     |                        | 2024-05-17 23:40:13 | 1                  |

☐ Check all | With selected: Edit Copy Delete Export

Figure ?? Table interview\_feedback

Server: 127.0.0.1 » Database: fypzakat2 » Table: notifications

Showing rows 0 - 0 (1 total, Query took 0.0003 seconds.)

`SELECT * FROM `notifications``

☐ Profiling [ Edit inline ] [ Edit ]

☐ Show all | Number of rows: 25 | Filter rows: Search this table

Extra options

|                                           | id | user_id | message                                               | is_read | created_at          |
|-------------------------------------------|----|---------|-------------------------------------------------------|---------|---------------------|
| <input type="checkbox"/> Edit Copy Delete | 29 | 40      | Hi! "Greetings from ZakatEase! We extend a warm in... | 0       | 2024-06-19 16:21:41 |

☐ Check all | With selected: Edit Copy Delete Export

Figure ?? Table notifications

Server: 127.0.0.1 » Database: fypzakat2 » Table: emails

☐ Profiling [ Edit inline ] [ Edit ] [ Explain SQL ] [ Create PHP code ] [ Refresh ]

☐ Show all | Number of rows: 25 | Filter rows: Search this table | Sort by key: None

Extra options

|                                           | id | user_id | email                   | subject                       | message                                               | sent_at             |
|-------------------------------------------|----|---------|-------------------------|-------------------------------|-------------------------------------------------------|---------------------|
| <input type="checkbox"/> Edit Copy Delete | 9  | 33      | alyashada1212@gmail.com | Your zakat Status             | Hi! &#34;Greetings from ZakatEase! We extend a war... | 2024-06-17 22:59:38 |
| <input type="checkbox"/> Edit Copy Delete | 10 | 33      | alyashada1212@gmail.com | Your zakat Status             | Hi! &#34;Greetings from ZakatEase! We extend a war... | 2024-06-17 23:02:10 |
| <input type="checkbox"/> Edit Copy Delete | 11 | 33      | alyashada1212@gmail.com | ZakatEase - Your zakat Status | Hi! &#34;Greetings from ZakatEase! We extend a war... | 2024-06-17 23:15:41 |
| <input type="checkbox"/> Edit Copy Delete | 13 | 33      | alyashada1212@gmail.com | ZakatEase - Your zakat Status | Hi! &#34;Greetings from ZakatEase! We extend a war... | 2024-06-18 23:14:22 |

☐ Check all | With selected: Edit Copy Delete Export

Figure ?? Table emails

Server: 127.0.0.1 » Database: fypzakat2 » Table: zakat\_application\_dates

☐ Profiling [ Edit inline ] [ Edit ] [ Explain SQL ] [ Create PHP code ] [ Refresh ]

☐ Show all | Number of rows: 25 | Filter rows: Search this table

Extra options

|                                           | id | start_date | end_date   |
|-------------------------------------------|----|------------|------------|
| <input type="checkbox"/> Edit Copy Delete | 1  | 2024-06-08 | 2024-07-08 |

☐ Check all | With selected: Edit Copy Delete Export

**Figure ??** Table zakat\_application\_dates

## **4.5 Interface Design**

This section discusses the interface design of web-based system which include the screenshot of login page, sign up page, home page, dashboard page, profile page, notification page, calculator page, zakat application form page (student), assigned application page (interviewer), manage user page (admin), manage application page (admin), set open/close application date page (admin).

### **4.5.1 Signup Page**

The signup page is where the user needs to sign up for account and set up login credentials which user need to fill in the information regarding the user type whether student, admin or interviewer, full name, email address, username and password. The signup page allow user to gain access to the web-based system.

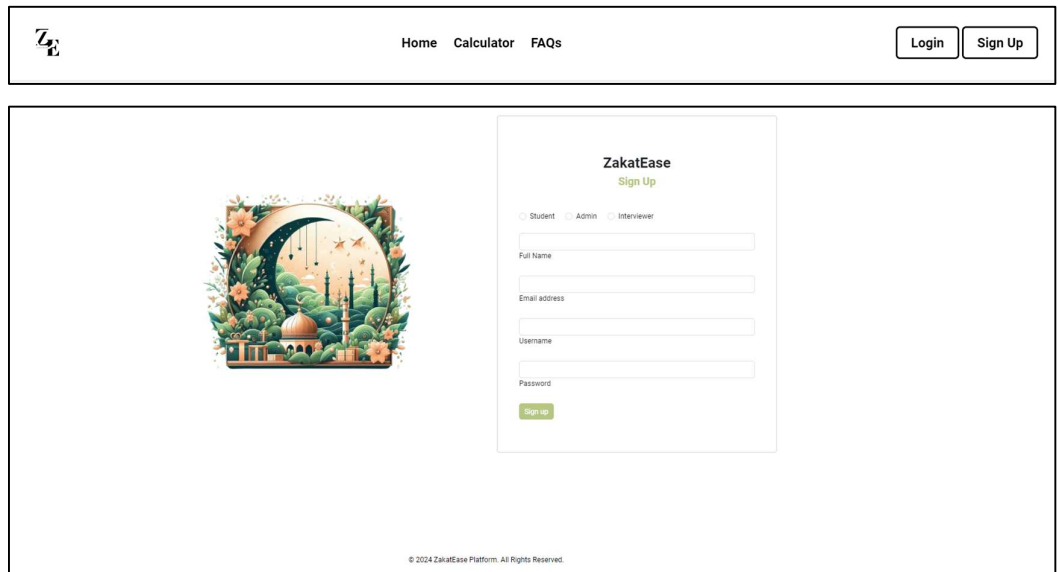


Figure ?? Signup page

#### 4.5.2 Login Page

After the signup process, user will need to login to the account. Users are required to fill in the role, username, and password. The login page is an essential part of our application, providing a secure interface for users to access their accounts. The login page serve as authentication to verifies the identity of user by checking the credentials.

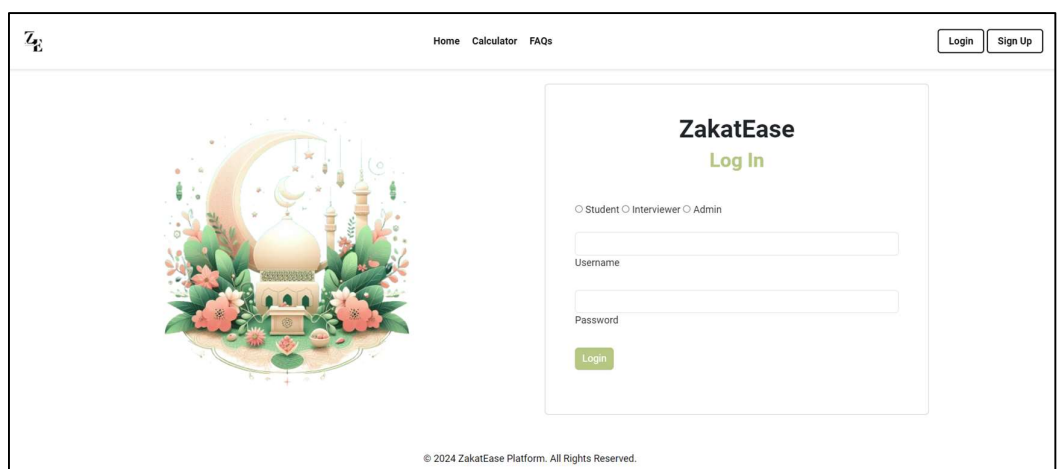
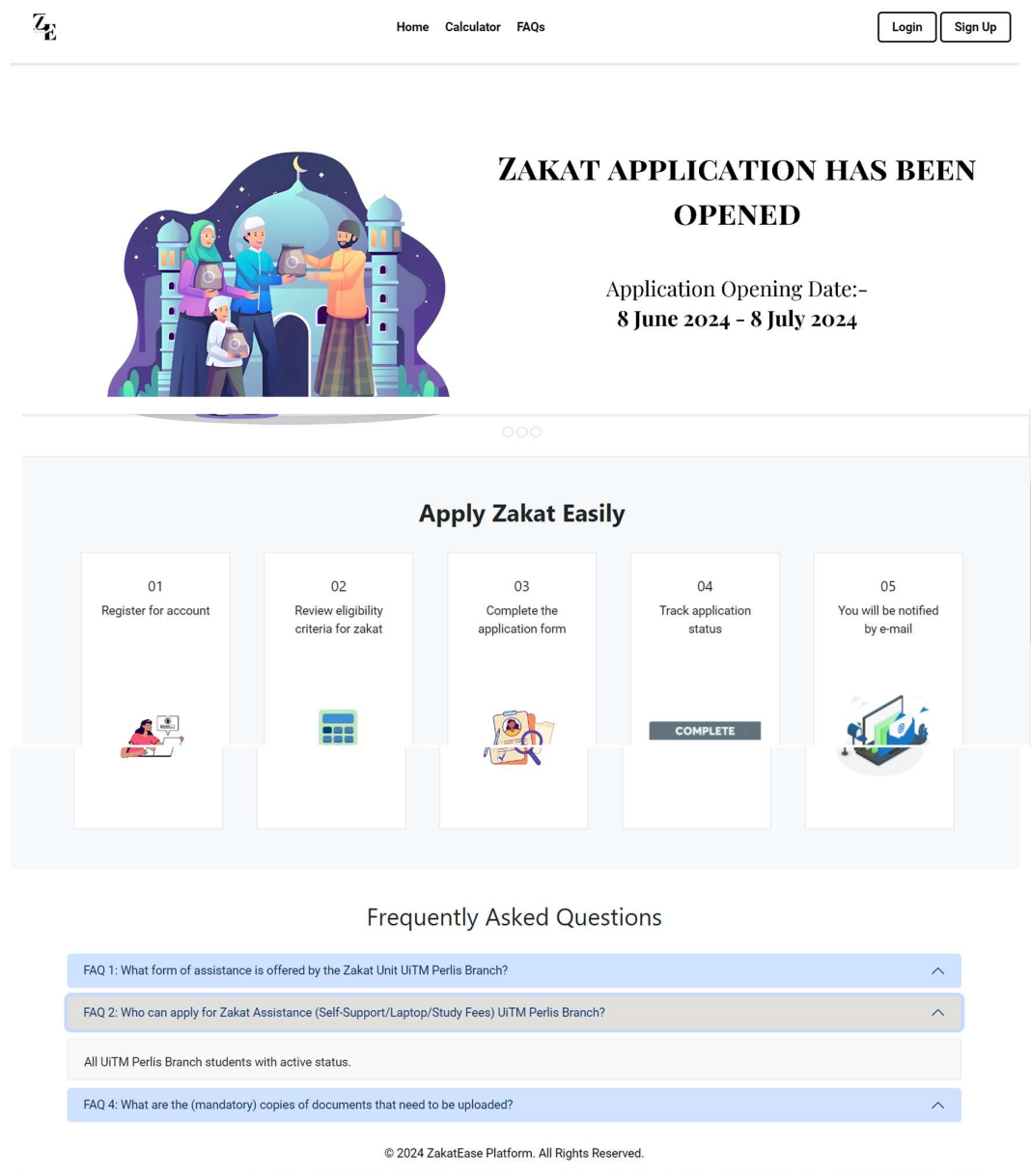


Figure ?? Login page

### 4.5.3 Home Page

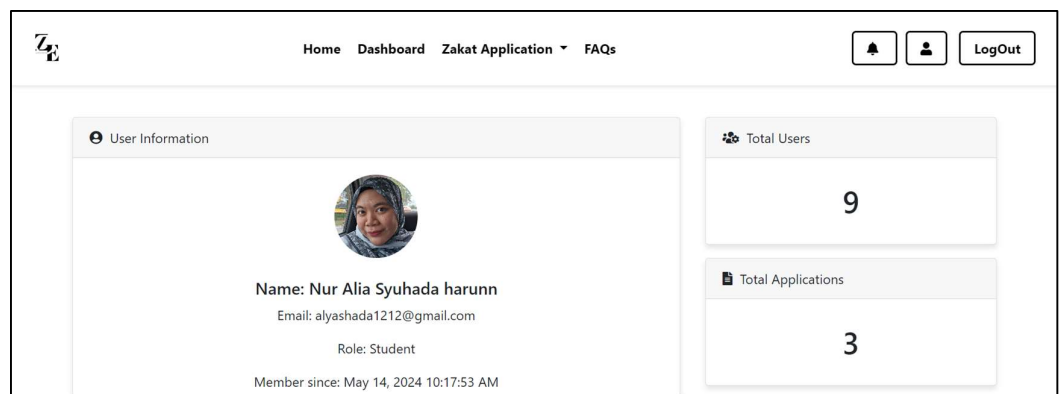
The home page is designed to deliver the important information related to zakat application. It is designed to give users with an easy and entertaining experience, allowing them to readily access the information. The home page includes section related to latest information. Then, there are section that provides information on how user can apply the zakat and frequently asked questions.



**Figure ?? Home page**

#### **4.5.4 Dashboard Page**

The dashboard page presenting the information related to user in clear and organized manner which provide clarity. The dashboard page provides users with overview of the account which include the user information such as name, email, role, and date user join the platform. The other related information is the total of users, and application that has been made.



**Figure ?? Dashboard page**

#### **4.5.5 Profile Page**

The profile page is an easy-to-use interface for updating and managing personal information such as profile images, full names, mobile numbers, and addresses. This feature allows users to save correct and up-to-date information, which improves the entire experience on the site. The profile page develops a sense of identification and belonging by allowing users to add photographs and correct names to their profiles.

**Figure ??** Profile page

#### 4.5.6 Notification Page

The notification page is intended to give notification related to the zakat application after user have made an application. Therefore, after the application has been evaluated by responsible authorities which are admin and interviewer, student would be given notification that include unique QR code on this page. The QR code need to be scan for user to know about the zakat status of zakat application. The QR code as would include the related details regarding the interview session if they got successful to go to interview.

**Figure ??** Notification page

### 4.5.7 Calculator Page

The calculator page provide user with calculator functionality to determine whether the user is eligible for applying for zakat or not. User can input the financial information including Total Sponsorship/Semester and Other Income. The calculator also considers the constant Had Kifayah amount of RM 3403.00, which is a standard threshold. This functionality clarifies users' eligibility status and assures correctness during the zakat application procedure. Users can use this tool with confidence to determine their eligibility for zakat and proceed with their application.

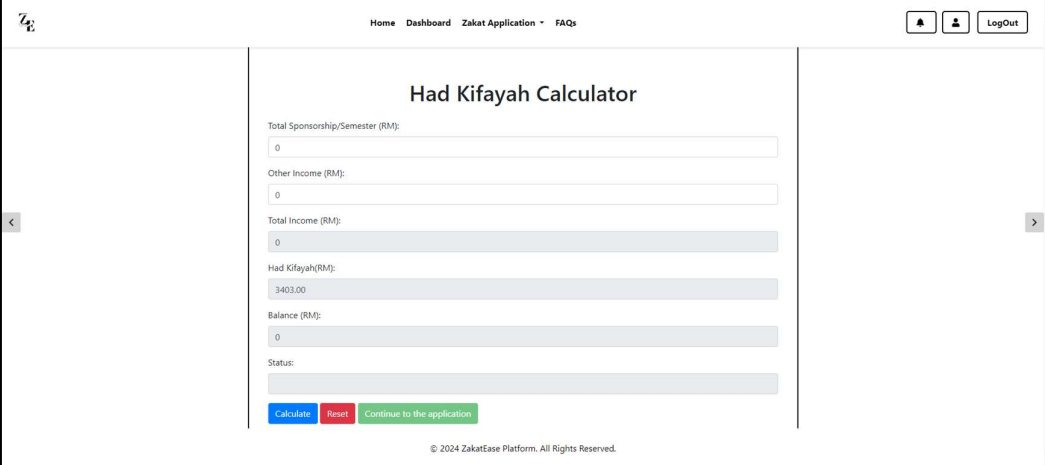


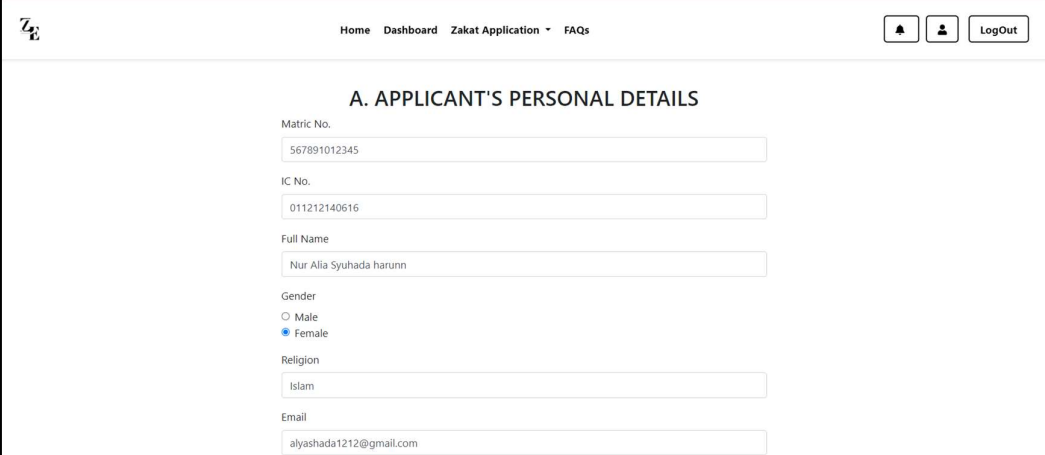
Figure ?? Calculator page

### 4.5.8 Zakat Application Form Page (Student)

The form has numerous sections: A. Applicant's Personal Details, where users provide their basic personal information; B. Family Background (Parents/Guardian), which gathers details about the applicant's family; C. Dependency of the Head of the Family (Parent/Guardian), which assesses the financial dependents of the family head; D. Siblings Information (Including the Applicant), to understand the family structure; E. Financial Information, where applicants detail their income sources and financial status. This



organised form guarantees that all necessary information is gathered to allow for informed choices on zakat eligibility and distribution.



The screenshot displays a web application interface for a student's zakat application. At the top, there is a navigation bar with a logo on the left and links for 'Home', 'Dashboard', 'Zakat Application', and 'FAQs' in the center. On the right side of the navigation bar are three icons: a bell, a user profile, and a 'LogOut' button. Below the navigation bar, the main content area is titled 'A. APPLICANT'S PERSONAL DETAILS'. This section contains several form fields: 'Matric No.' with the value '567891012345', 'IC No.' with '011212140616', 'Full Name' with 'Nur Alia Syuhada harunn', 'Gender' with radio buttons for 'Male' and 'Female' (where 'Female' is selected), 'Religion' with 'Islam', and 'Email' with 'alyashada1212@gmail.com'.

**Figure??** Zakat Application Form page (Student)

#### 4.5.9 Assigned Application Page (Interviewer)

The assigned application page can be accessed by the interviewer, providing the interviewer with the list of applications that have been assigned to them by the admin. There is a button that allows the interviewer to view the details of each application. Moreover, the interviewer can generate and assign a QR code to set up the interview details, which include the interview date, interview time, interview location, and additional notes. The interviewer will need to send the notification and email, which include the status of the applicant's interview details. Therefore, the applicant needs to scan the QR code to know the status and details of their application. After conducting the interview session, the interviewer has the ability to fill in the feedback form regarding the interview session, which will indicate the interviewer's response to the interview session. The feedback then will be accessible to the admin, who will review the feedback and make an informed decision to further the application in the meeting with the responsible authorities. This system streamlines the interview process, ensures clear communication, and facilitates efficient decision-making.

ZakatEase

DashboardAssigned Application

Logout

Applications Assigned to You

| Application ID | Applicant Name          | Matric No    | Interviewer Name | Application                 | Interview Details       | Assign QR Code              | QR Code                                                                                               | Send Email & Notification | Feedback           | Interview Status    |
|----------------|-------------------------|--------------|------------------|-----------------------------|-------------------------|-----------------------------|-------------------------------------------------------------------------------------------------------|---------------------------|--------------------|---------------------|
| 42             | Nur Alia Syuhada harunn | 567891012345 | MethKiller       | <div>View Application</div> | <div>View Details</div> | <div>Generate QR Code</div> | <div>QR Code</div> |                           | <div>Actions</div> | <div>Feedback</div> |
| 43             | Aina Abdul              | 5464         | MethKiller       | <div>View Application</div> | <div>View Details</div> | <div>Generate QR Code</div> | <div></div>        |                           | <div>Actions</div> | <div>Feedback</div> |
| 44             | haziq                   | 34           | MethKiller       | <div>View Application</div> | <div>View Details</div> | <div>Generate QR Code</div> | <div>QR Code</div> |                           | <div>Actions</div> | <div>Feedback</div> |

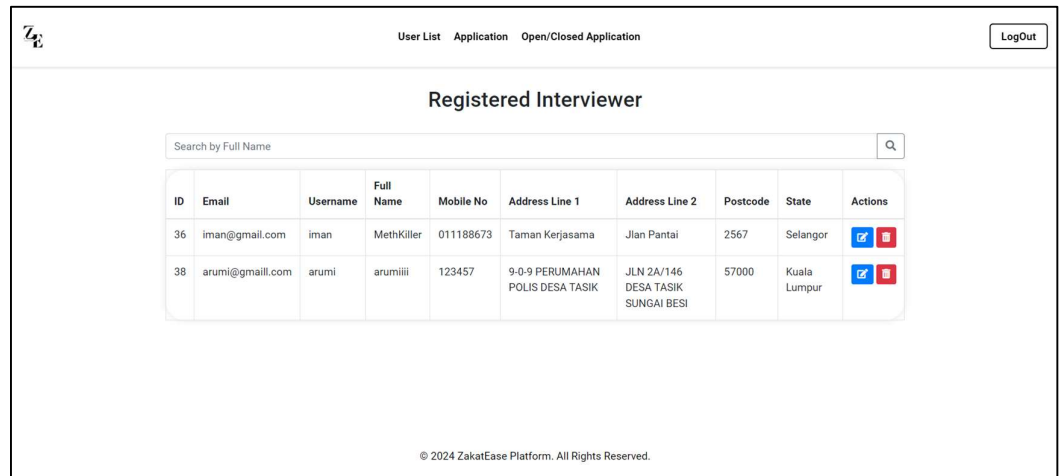
Figure?? Assigned Application Page (Interviewer)

4.5.10 Manage User Page (Admin)

The manage page user enable admin to efficiently manage the information of registered users. This page allow admin to make changes and delete the information as needed, ensuring that the information is accurate and up to date. This capability enables administrators to efficiently manage user profiles, so guaranteeing that the platform runs smoothly and that all user records are current and valid.

| <div>  <div>User List Application Open/Closed Application</div> <div>  <div>Logout</div> </div> </div> |                         |          |                         |              |                                       |                                   |          |              |                                                                                                                                                                             |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------|----------|-------------------------|--------------|---------------------------------------|-----------------------------------|----------|--------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Registered Students                                                                                                                                                                                                                                                            |                         |          |                         |              |                                       |                                   |          |              |                                                                                                                                                                             |
| <div> <div>Search by full name</div> <div></div> </div>                                                                                                                                   |                         |          |                         |              |                                       |                                   |          |              |                                                                                                                                                                             |
| ID                                                                                                                                                                                                                                                                             | Email                   | Username | Full Name               | Mobile No    | Address Line 1                        | Address Line 2                    | Postcode | State        | Actions                                                                                                                                                                     |
| 33                                                                                                                                                                                                                                                                             | alyashada1212@gmail.com | alia121  | Nur Alia Syuhada harunn | 011257063132 | BLOK 9-0-9 PERUMAHAN POLIS DESA TASIK | SUNGAJ BESI 57000 KUALA LUMPUR    | 57000    | Kuala Lumpur |   |
| 34                                                                                                                                                                                                                                                                             | aina@gmail.com          | aina     | Aina Abdul              | 34           | 9-0-9 PERUMAHAN POLIS DESA TASIK      | JLN 2A/146 DESA TASIK SUNGAJ BESI | 57000    | Kuala Lumpur |   |

Figure?? Manage User Page (Admin) - Student



**Figure??** Manage User Page (Admin) - Interviewer

#### 4.5.11 Manage Application Page (Admin)

The Manage Application Page is an essential tool for administrators, offering extensive access to all submitted applications. This page allows administrators to access extensive information about each application, ensuring complete examination and processing. Admins can change the status of applications to reject, approve, or pending based on information from application. Additionally, the page allows administrators to analyse the comments offered by interviewees, which is critical for making educated choices regarding each applicant's eligibility for Zakat. This tool enables a simplified and transparent application management procedure, allowing efficient and effective administration of zakat distribution.

</

**Figure?? Manage Application Page (Admin)**

#### 4.5.12 Set Open/close Application Date Page (Admin)

The set open/close application date is page that allows administrators to set the opening and closing dates for zakat applications. Through this form, administrators may set the time range during which users can submit applications. Once these dates are chosen, users will only be able to view the application form within the specified period. This guarantees that the application process is well-organized and follows a set timetable, resulting in improved administration and timely processing of all applications. By regulating the application window, administrators may efficiently manage the flow of applications and guarantee that all submissions are received and processed on time.

The screenshot shows a web application interface for an administrator. At the top, there is a navigation bar with a logo on the left, three menu items: 'User List', 'Application', and 'Open/Closed Application', and a 'LogOut' button on the right. The main content area is titled 'Update Zakat Application Dates'. Below the title, there are two date input fields. The first is labeled 'Start Date' and contains the text '08/06/2024'. The second is labeled 'End Date' and contains the text '08/07/2024'. Both input fields have a small square icon on the right side. Below these fields is a blue button with the text 'Update Dates'.

**Figure??** Set Open/closed Application Date Page (Admin)

## 4.6 Summary

This chapter discusses web-based application development, including the interface, software used, and database utilization. The interface consists of a page the user can navigate to fulfil the functional requirements. Non-functional requirements are considered as well. The database section shows how the table was managed and explains how the Entity Relationship Diagram (ERD) works. The QR code integration was discussed, as well as how the user could use the function of the QR Code in the web-based application. Therefore, from all the developments that have been made, the first objective, which is to develop a web-based system of zakat application with QR code integration, was accomplished.